

4 (**short**) As shown in Fig. 4, helium gas of mass 0.075 kg is maintained at a constant pressure $p = 5$ bar by means of a weighted frictionless piston in a cylinder. It may be assumed that helium behaves as a perfect gas.

(a) In its initial state, the helium occupies a volume of 0.5 m^3 . Calculate the initial temperature of the gas. [3]

(b) The helium is cooled until its temperature reaches 0°C . Taking the helium as the thermodynamic system, calculate the work done by the system, the change in internal energy of the system, and the heat transferred to the system, during the cooling process. [7]

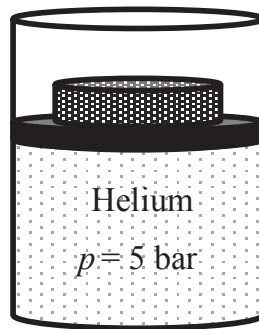


Fig. 4